ARDUINO Simplified Cheat Sheet

Structures

Program Structure
```
void setup() {
  // runs once when sketch starts
}
void loop() {
  // runs repeatedly
}
```

Control Structures
```
if (condition) { ... } else { ... }

while (condition) { ... }
for (initialization; condition;
  increment/decrement) { ... }

break;    // exit from loop immediately

continue;  // go to next iteration

switch (expression) {
  case 1: ...
    break;
  case 2: ...
    break;
  default:...
}
```

Function Definition
```
returnType functionName(parameters...)
{
  ...
  return value;
}
```

Operators

General Operators

=	assignment		
+	addition	-	subtraction
*	multiplication	/	division
%	modulus		
==	equality	!=	inequality
<	less than	>	greater than
<=	less than or equal		
>=	greater than or equal		
&&	logical AND	||	logical OR
!	logical NOT		

Compound Operators

++	increment
--	decrement
+=	addition and assignment
-=	subtraction and assignment
*=	multiplication and assignment
/=	division and assignment
%=	modulus and assignment
&=	bitwise AND and assignment
^=	bitwise XOR and assignment
|=	bitwise OR and assignment

Bitwise Operators

&	bitwise AND	|	bitwise OR
^	bitwise XOR	~	bitwise NOT
<<	left shift	>>	right shift

Pointer Access

&	address of
*	value at

Built-in Functions

Pin Input/Output
```
Digital I/O - pins 0-13 and A0-A5
  pinMode(pin,
    [INPUT, OUTPUT, INPUT_PULLUP])
int digitalRead(pin)
digitalWrite(pin, [HIGH, LOW])

Analog In - pins A0-A5
  int analogRead(pin)
  analogReference(
    [DEFAULT, INTERNAL, EXTERNAL])

PWM - pins 3, 5, 6, 9, 10 and 11
  analogWrite(pin, value)
```

Advanced I/O
```
tone(pin, frequency in Hz)
tone(pin, frequency in Hz, duration
  in ms)
noTone(pin)
shiftOut(dataPin, clockPin,
  [MSBFIRST, LSBFIRST], value)
unsigned long pulseIn(pin,
  [HIGH, LOW])
```

Time
```
unsigned long millis()
  // overflows after 50 days
unsigned long micros()
  // overflows after 70 minutes
delay(ms)
delayMicroseconds(us)
```

Math
```
min(x, y)   max(x, y)   abs(x)
sin(rad)    cos(rad)    tan(rad)
sqrt(x)     pow(base, exponent)
constrain(x, minval, maxval)
map(val, fromL, fromH, toL, toH)
```

Random Numbers
```
randomSeed(seed) // long or int
long random(max) // 0 to max-1
long random(min, max)
```

Bits and Bytes
```
lowByte(x)        highByte(x)
bitRead(x, bitn)
bitWrite(x, bitn, bit)
bitSet(x, bitn)   bitClear(x, bitn)
bit(bitn)  // bitn: 0=LSB 7=MSB
```

Type Conversions
```
char(val)       byte(val)
int(val)        word(val)
long(val)       float(val)
```

External Interrupts
```
int interrupt number
  digitalPinToInterrupt(pin)
attachInterrupt(interrupt, handler,
  [LOW, CHANGE, RISING, FALLING])
detachInterrupt(interrupt number)
interrupts()    // enable
noInterrupts()  // disable
```

Libraries

Serial communicate via RX/TX
```
begin(long baud rate) // up to 115200
end()
int available()  // bytes available
int read()
int peek()  // read without removing
flush()
print(data)    println(data)
write(byte)    write(char *string)
write(byte *data, size)
SerialEvent() // called if data ready
```

SoftwareSerial.h comm. on any pin
```
SoftwareSerial(rxPin, txPin)
begin(long baud rate) // up to 115200
listen()// enable port for listening
isListening() // test port
read(), peek(), print(), println(),
write() // same as of Serial library
```

EEPROM.h access EEPROM byte
```
read(int address)
write(int address, byte data)
EEPROM[index]  // access as array
```

Servo.h control servo motors
```
attach(pin, [min_uS, max_uS])
write(angle)  // 0 to 180
writeMicroseconds(uS)
  // 1000 to 2000 via 1500 midpoint
int read()    // 0 to 180
bool attached() // is attached
detach()
```

Wire.h I²C communication
```
begin()     // join a master
begin(address) // join a slave
requestFrom(address, count)
beginTransmission(address) // step 1
send(byte)              // step 2
send(char * string)
send(byte * data, size)
endTransmission()       // step 3
int available() // bytes available
byte receive()  // get next byte
onReceive(handler to call)
onRequest(handler to call)
```

Data, Literals, Variables and Arrays

Data Types

boolean	true | false
char	-128 - 127
unsigned char	0 - 255
byte	0 - 255
int	-32768 - 32767
unsigned int	0 - 65535
word	0 - 65535
long	-2147483648 - 2147483647
unsigned long	0 - 4294967295
float	-3.4028e+38 - 3.4028e+38
double	same as float
void	no value

Strings
```
char str1[8] =
  {'A','r','d','u','i','n','o','\0'};
  // includes \0 null termination
char str2[8] =
  {'A','r','d','u','i','n','o'};
   // compiler adds null termination
char str3[] = "Arduino";
char str4[8] = "Arduino";
```

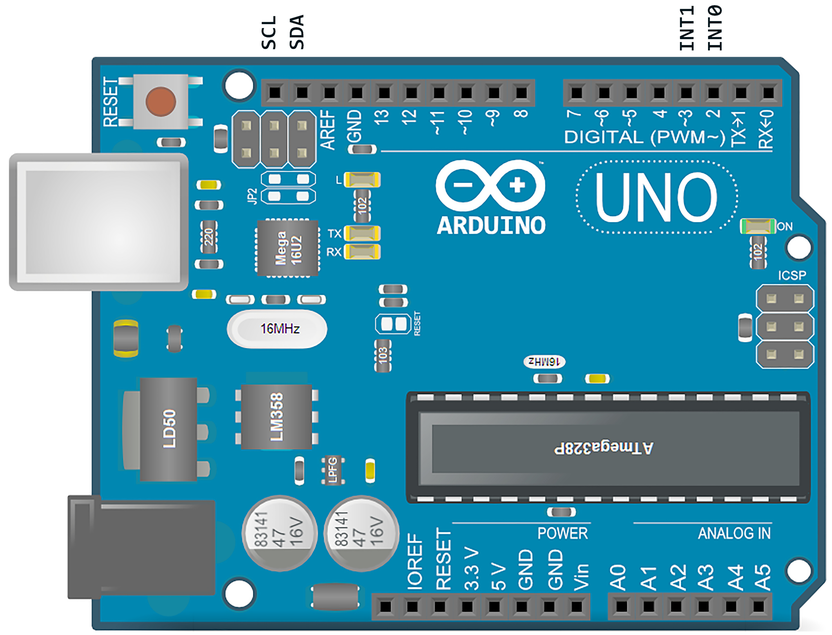
Numeric Literals

123	decimal
0b01111011	binary
0173	octal
0x7B	hexadecimal
123U	unsigned integer
123L	long
123UL	unsigned long
123.0	floating point
1.23e6	$1.23x10^6$ = 1230000

Qualifiers

static	persists between calls
volatile	unpredictable
const	read-only
PROGMEM	in flash instead of RAM

Arrays
```
type arrayName[size];
type arrayName[size] = {val1,
  val2, ...};
type arrayName[] = {val1, val2, ...};
arryName[index] // accessing
```



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